THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 70	Area $= \frac{1}{2} \times a \times b$.
2	**(a)** 25	Intercepts are c and $-c/m$; area $= \frac{1}{2}\lvert c\rvert \cdot \lvert c/m\rvert$.
3	**(b)** $\frac{7}{9}$	Each is $\frac{d}{9}$.
4	**(b)** 35	$x^3+y^3=(x+y)^3-3xy(x+y)$.
5	**(d)** 31	Add the two distances from the centre.
6	**(d)** 96	Use Brahmagupta's formula.
7	**(a)** $\frac{1}{9}$	List all pairs with the given product.
8	**(b)** 50	$d=3$, $a=5$; $t_{16}=a+15d$.
9	**(c)** 21	Squares have an odd number of factors; count the integers whose squares lie in the range.
10	**(d)** 187	$34(78-111)=n_2(72-78)$.
11	**(c)** 10	Trace the sequence step by step.
12	**(a)** 15°	Angles $a, a+d, a+2d, a+3d$ add up to 360°.
13	**(d)** 153	Intercepts 18 and 17; area $= \frac{1}{2} \times 18 \times 17$.
14	**(d)** 36 cm	$\sqrt{l^2+b^2+h^2}=\sqrt{144+576+576}$.
15	**(b)** 5	The circumcentre is the midpoint of the hypotenuse; its distance from the right-angle vertex is half the hypotenuse.
16	**(b)** 4	Equate the two expressions.
17	**(a)** Both A and R are true and R is the correct explanation of A.	$x=-3<0$ and $y=4>0$, which is exactly the sign pattern of Quadrant II.
18	**(a)** Both A and R are true and R is the correct explanation of A.	$-\dfrac{-12}{3}=4$; R is the general rule that gives it.
19	**(a)** Both A and R are true and R is the correct explanation of A.	$0.375=\dfrac{375}{1000}=\dfrac{3}{8}$; R is the general reason.
20	**(a)** Both A and R are true and R is the correct explanation of A.	$103^2=(100+3)^2=10000+600+9=10609$, using R.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Circumcentre is the midpoint of $AB=(3,4)$ (1); Radius $=OM=\sqrt{9+16}=5$ (1).	1+1
22	$200+30x=440+10x$ (1); $x=12$ units (1).	1+1
23	$0.\overline{17}=\frac{17}{99}$, $0.\overline{71}=\frac{71}{99}$ (1); Sum $=\frac{8}{9}$ (1).	1+1
24	$x^3+y^3=(x+y)^3-3xy(x+y)$ (1); $=512-288=224$ (1).	1+1
25	Distances: $\sqrt{25^2-7^2}=24$ cm and $\sqrt{25^2-24^2}=7$ cm (1); Distance $=24+7=31$ cm (1).	1+1
26	$s=23$ cm (1); $A=\sqrt{(s-a)(s-b)(s-c)(s-d)}=96$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Let the point be $(x,0)$: $(x-2)^2+400=625$ (1); $(x-2)^2=225$ (1); $x=17$ or $x=-13$ (1).	1+1+1
28	$5x+2=2x+11$ (1); $x=3$ (1); $y=17$ (1).	1+1+1
29	$x=0.4\overline{7}$: $100x-10x=43$ (1); $x=\frac{43}{90}$ (1); Long division of p by q gives $0.4\overline{7}$ (1).	1+1+1
30	$\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=7^3-3(7)$ (1); $=322$ (1).	1+1+1
31	$d_1=\sqrt{13^2-12^2}=5$ cm, $d_2=\sqrt{15^2-12^2}=9$ cm (1); Opposite sides: $5+9=14$ cm (1); Same side: $\lvert 5-9\rvert=4$ cm (1).	1+1+1

	OR: Radius $= \frac{44}{2\pi} = 7$ cm; Side $= \frac{44}{4} = 11$ cm (1); Circle area $= 154$ cm^2; Square area $= 121$ cm^2 (1); The circle encloses the larger area (1).	1+1+1
32	$P(\text{green}) = 1 - \frac{1+1}{3} = \frac{1}{3}$; Red $= 8$ (1); Blue $= 8$ (1); Green $= 8$ (1).	1+1+1
	OR: $2a + 10d = 40$ and $a + 4d = 17$ (1); Solving: $a = 5$, $d = 3$ (1); $S_{20} = 670$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $x < 0,\ y > 0$, so Quadrant II (1). (ii) $(-3, -4)$ (1). (iii) Same abscissa, so $AB = \lvert 4 - (-2) \rvert = 6$ km (1). (iv) Midpoint $= \left(\frac{-3+5}{2}, \frac{4-2}{2} \right) = (1, 1)$ (1); $AC = \sqrt{8^2 + 6^2} = 10$ km (1).	1+1+1+2
34	(i) $C(x) = 12x + 50$ (1). (ii) $C(10) = 120 + 50 =$ Rs 170 (1). (iii) $12x + 50 = 230 \Rightarrow x = 15$ km (1). (iv) Slope $= 12$: the rate of Rs 12 per km (1); y-intercept $= 50$: the fixed charge of Rs 50 (1).	1+1+1+2
35	Divide 3 by 7 until a remainder repeats (1); $\frac{3}{7} = 0.\overline{428571}$ (1); Block 428571 (1); Block length $= 6$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
36	$x^2 + y^2 = 9 + 56 = 65$ (1); $x^3 - y^3 = (x-y)^3 + 3xy(x-y) = 27 + 252$ (1); $x^3 - y^3 = 279$ (1); $(x+y)^2 = (x-y)^2 + 4xy = 9 + 112$ (1); $(x+y)^2 = 121$ (1).	1+1+1+1+1
	OR: $\angle A + \angle C = 180^\circ$ (1); $5x = 150$, so $x = 30$ (1); $\angle B + \angle D = 180^\circ$ (1); $6y = 168$, so $y = 28$ (1); $\angle A = 70^\circ$, $\angle C = 110^\circ$, $\angle B = 120^\circ$, $\angle D = 60^\circ$ (1).	1+1+1+1+1
37	$AC = \sqrt{6^2 + 8^2} = 10$ cm (1); Area $\triangle ABC = \frac{1}{2} \times 6 \times 8 = 24$ cm^2 (1); $\triangle ACD$ has sides 10, 17, 21 cm: $s = 24$ cm (1); Area $\triangle ACD = 84$ cm^2 (1); Area $ABCD = 108$ cm^2 (1).	1+1+1+1+1
	OR: Sum 6: 5 outcomes; more than 6: 21 (1); $P(\text{sum} = 6) = \frac{5}{36}$ (1); $P(> 6) = \frac{7}{12}$ (1); $P(\leq 6) = 1 - \frac{7}{12} = \frac{5}{12}$ (1); The two probabilities add to 1 (1).	1+1+1+1+1
38	Penalties form an AP (1); $t_{30} = 200 + 29 \times 50 = 1650$ (1); $S_n = \frac{n}{2}(a + l)$ (1); $S_{30} = \frac{30}{2}(200 + 1650)$ (1); Total ₹27750 (1).	1+1+1+1+1
	OR: $5 \times 6 \times 7 = 210 = 35 \times 6$ (1); Among three consecutive integers one is a multiple of 3 (1); At least one is even (1); Both factors 2 and 3 present, so divisible by 6 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{2}{9}$	Count the favourable pairs out of 36.
2	**(c)** 37	$t_k = S_k - S_{k-1} = p(2k-1)+q$.
3	**(a)** 11	$n = 11$ gives 121, which is composite.
4	**(b)** 7 : 9	Amounts are in the ratio $(41-34) : (34-25) = 7 : 9$.
5	**(d)** 14	Pairs: $(1344, 872) \to (872, 472) \to (472, 400) \to (400, 72) \to (328, 72) \to (256, 72) \to (184, 72) \to (112, 72) \to (72, 40) \to (40, 32) \to (32, 8) \to (24, 8) \to (16, 8) \to (8, 8) \to (8, 0)$.
6	**(b)** 70 cm	Sides 15 cm and 20 cm.
7	**(b)** 1101	Rate $= \dfrac{182}{7} = 26$ per km; fixed charge 113.
8	**(b)** 22176 cm^3	Volume of hemisphere + volume of cone.
9	**(d)** 41	Radius $= \frac{1}{2} \times$ distance between the endpoints.
10	**(c)** 25	Evaluate the inner value first.
11	**(d)** 3	Cancel every prime factor other than 2 and 5.
12	**(b)** 18	$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$.
13	**(a)** 14	Add the distances of the centres from the common chord.
14	**(d)** 770	Difference of the two areas.
15	**(c)** $\frac{7}{8}$	Use the complement rule where it is quicker.
16	**(c)** 816	AP with $a = 6$, $d = 6$.
17	**(a)** Both A and R are true and R is the correct explanation of A.	Half-chord $= \sqrt{5^2 - 3^2} = 4$ cm, using the perpendicular bisection in R; chord $= 8$ cm.
18	**(a)** Both A and R are true and R is the correct explanation of A.	$s = 21$, area $= \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$, using R.
19	**(a)** Both A and R are true and R is the correct explanation of A.	$P = \dfrac{3}{6} = \dfrac{1}{2}$, obtained exactly as in R.
20	**(a)** Both A and R are true and R is the correct explanation of A.	$5 - 2 = 8 - 5 = 11 - 8 = 3$; a constant common difference is the definition of an AP.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Favourable pairs: (2,6), (3,4), (4,3), (6,2) (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1
22	$\frac{n}{2}[2a + (n-1)d] = 255$; $d = 2$ (1); $t_{15} = 31$ (1).	1+1
23	$x^2 = 20^2 + 21^2 = 841$ (1); $x = 29$ (1).	1+1
24	Let amounts be x and y; $16x + 64y = 36(x + y)$ (1); $20x = 28y$; $x : y = 7 : 5$ (1).	1+1
25	gcd(380, 520): $(520, 380) \to (380, 140) \to (240, 140) \to (140, 100) \to (100, 40) \to (60, 40) \to (40, 20) \to (20, 20) \to (20, 0)$ giving 20; gcd(20, 152): $(152, 20) \to (132, 20) \to (112, 20) \to (92, 20) \to (72, 20) \to (52, 20) \to (32, 20) \to (20, 12) \to (12, 8) \to (8, 4) \to (4, 4) \to (4, 0)$ (1); gcd of the three numbers $= 4$ (1).	1+1
26	Half of the given diagonal $= 24$ cm; Half of the other diagonal $= \sqrt{51^2 - 24^2} = 45$ cm, so the diagonal is 90 cm (1); Area $= \frac{1}{2} \times 48 \times 90 = 2160$ cm^2; Height $= \dfrac{\text{area}}{\text{side}} = \dfrac{2160}{51} = 42.35$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$8^2 + 9^2 = 64 + 81 = 145$, odd; General: $n^2 + (n+1)^2 = 2n^2 + 2n + 1$ (1); $= 2(n^2 + n) + 1$ (1); Odd (1).	1+1+1
28	Let the number of remaining students be n; $40 \times 49 + 90n = 70(40 + n)$ (1); $n(90 - 70) = 40(70 - 49)$ (1); $n = 42$ (1).	1+1+1

29	For $i = 1$ the inner loop runs 11 times, for $i = 2$ it runs 10 times, and so on (1); Count $= 11 + 10 + \cdots + 1 = \frac{11 \times 12}{2}$ (1); Count $= 66$ (1).	1+1+1
30	Diagonals bisect each other, so $OA = OC$; $5x + 3 = x + 147$ gives $x = 36$ (1); $OA = 183$ cm (1); $AC = 2 \times 183 = 366$ cm (1).	1+1+1
31	Slope $= \dfrac{17 - 0}{0 - 18} = -\frac{17}{18}$; Equation: $y = -\frac{17}{18}x + 17$, i.e. $17x + 18y = 306$ (1); Check: (18, 0)$gives$17\times18=306 (1); $Area$=\frac12\times18\times17=153$ square units (1).	1+1+1
	OR: $\frac{4}{3}\pi 12^3 = \pi(3)^2 h$ (1); $h = \dfrac{4 \times 1728}{3 \times 9} = 256$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 3 \times 256 = 4827.43$ cm^2 (1).	1+1+1
32	$C = B + D - A = (-7, 24)$ (1); $AB = 15$, $AD = 20$ (1); $AC = \sqrt{225 + 400} = 25$ (1).	1+1+1
	OR: $p(3) - p(1) = 2a = 8$, so $a = 4$; $b = 8$, $p(x) = 4x + 8$ (1); Zero $x = -2$ (1); $p(6) = 32$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $\dfrac{7}{16} = 0.4375$ (1). (ii) $\dfrac{11}{25} = 0.44$ (1). (iii) $12 = 2^2 \times 3$ (1). (iv) $\frac{7}{16}$ and $\frac{11}{25}$ terminate (1) because $16 = 2^4$ and $25 = 5^2$ have only the prime factors 2 and 5; $\dfrac{5}{12} = 0.41\overline{6}$ does not terminate because 12 has the factor 3 (1).	1+1+1+2
34	(i) $(100 + 3)^2 = 10000 + 600 + 9 = 10609$ (1). (ii) $(100 - 2)^2 = 10000 - 400 + 4 = 9604$ (1). (iii) $(100 + 2)(100 - 2) = 10000 - 4 = 9996$ (1). (iv) $(1000 - 3)(1000 + 3) = 1000^2 - 3^2 = 999991$ (1) using $(a - b)(a + b) = a^2 - b^2$ (1).	1+1+1+2
35	Each side of the new figure is half a diagonal of the rectangle (midpoint theorem), and the diagonals of a rectangle are equal, so all four sides are equal (1); Diagonal of the rectangle $= \sqrt{196 + 2304} = 50$ cm, so each side $= 25$ cm (1); Perimeter $= 4 \times 25 = 100$ cm (1); Its diagonals are the sides of the rectangle, so area $= \frac{1}{2} \times 14 \times 48 = 336$ cm^2 (2).	1+1+1+2
36	Slope $= \dfrac{7 - (1)}{5 - (-1)} = 1$; $y = x + 2$ (1); $9 = 1k + 2$ gives $k = 7$ (2); x-intercept: $y = 0$ gives $x = -2$ (2).	1+2+2
	OR: Volume of the cone $= \frac{1}{3}\pi r^2 h = 103488$ cm^3 (1); Volume of the hemisphere $= \frac{2}{3}\pi r^3 = 155232$ cm^3 (1); Volume per cone $= 258720$ cm^3 (1); For 27 cones: $27 \times 258720 = 6985440$ cm^3 (2).	1+1+1+2
37	$PA^2 = PB^2$ with $P = (x, 0)$ (1); $x = 6$, so $P = (6, 0)$ (1); $PA^2 = 9 + 16 = 25$ (1); $PA = 5$ units (2).	1+1+1+2
	OR: Slope $= 3$ (1); Equation $y = 3x + 12$ (1); y-intercept 12, x-intercept -4 (1); Area $= \frac{1}{2} \times 12 \times 4$ (1); $= 24$ sq. units (1).	1+1+1+1+1
38	First $= 2(7 + 5) = 24$ (1); Second $= 7 - 5 = 2$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{35}$ cancel (2).	1+1+1+2
	OR: $2xy = (x + y)^2 - (x^2 + y^2) = 100 - 58 = 42$ (1); $xy = 21$ (1); $(x - y)^2 = x^2 + y^2 - 2xy = 58 - 42 = 16$ (1); $x - y = 4$ (1); $x^2 - y^2 = (x + y)(x - y) = 40$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 20	Equal slopes: $-\frac{4}{1} = -\frac{k}{5}$, so $k = 20$.
2	**(d)** 125	$\frac{(5)^3}{1^3} = 5^3$.
3	**(d)** $(0, 2)$	Set $PA = PB$ with $P = (0, y)$.
4	**(c)** 12	Equate $a_1 + b_1x = a_2 + b_2x$.
5	**(b)** 4	$\frac{1}{x} = 2 - \sqrt{3}$, so the sum is 2×2.
6	**(a)** 1030301	$101 = 100 + 1$; use $(a \pm b)^3$.
7	**(a)** 60°	Add the two parts of the angle.
8	**(d)** 35	Distance $= n \times 2\pi r$.
9	**(b)** 4	Let blue $= x$ and solve $\frac{x}{r+x} = \frac{p}{q}$.
10	**(b)** 7	$\frac{n}{2}[2a + (n-1)d] = 510$ gives d.
11	**(b)** 38	The middle number is the average, 36.
12	**(b)** 69	New salary is higher by $n \times 2 = 44$ thousand.
13	**(b)** 2075	The algorithm reverses the digits.
14	**(d)** 90 cm	Half-diagonals and the side form a right triangle: half of the other diagonal $= \sqrt{51^2 - 24^2} = 45$.
15	**(d)** 17	Equal slopes between consecutive points.
16	**(d)** 144	$\frac{54 \times 12 \times 6}{27}$.
17	**(a)** Both A and R are true and R is the correct explanation of A.	The converse is 'If a number is even, then it is divisible by 6'; 4 is a counterexample, which is exactly R.
18	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{20 \times 60 + 30 \times 70}{50} = \frac{3300}{50} = 66$, using R.
19	**(a)** Both A and R are true and R is the correct explanation of A.	$48 = 6 \times 8$ and $18 = 6 \times 3$, and no larger number divides both; R is the definition of gcd.
20	**(a)** Both A and R are true and R is the correct explanation of A.	$2 \times 180^\circ = 360^\circ$; R is the standard proof.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Substitute: $k(1) + 3(6) = 27$ gives $k = 9$; Line: $9x + 3y = 27$; $y = -3x + \frac{27}{3}$ (1); Slope $= -\frac{9}{3} = -3$; y-intercept $= \frac{27}{3}$ (1).	1+1
22	Diagonal $= \sqrt{6^2 + 18^2 + 27^2} = \sqrt{1089} = 33$ cm; TSA $= 2(108 + 486 + 162) = 1512$ cm^2 (1); Volume $= 6 \times 18 \times 27 = 2916$ cm^3 (1).	1+1
23	Centre $= (7, 9)$ (1); $AB = 2\sqrt{36 + 64} = 20$, radius $= 10$ (1).	1+1
24	$p(p(x)) = 3(3x + 4) + (4) = 9x + 16$ (1); Value $= 34$ (1).	1+1
25	$420 = 2^2 \times 3 \times 5 \times 7$ (1); Cancel the factors other than 2 and 5: $n = 21$ (1).	1+1
26	$103 = 100 + 3$ (1); $103^3 = 1092727$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$\frac{1}{x} = 2 - \sqrt{3}$ since $2^2 - 3 = 1$, so $x + \frac{1}{x} = 4$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 16 - 2 = 14$ (1).	1+1+1
28	Need two numbers with sum -5 and product 4; They are -1 and -4 (1); $= (x - 1)(x - 4)$ (1); At $x = 6$: $(5)(2) = 10$ (1).	1+1+1
29	Distances from the centre 12 cm and 5 cm (1); Same side: 7 cm (1); Opposite sides: 17 cm (1).	1+1+1

30	Boundary $= 2 \times 60 + 2\pi r$; $= 120 + 88 = 208$ m (1); Area $= 60 \times 28 + \pi r^2$ (1); $= 1680 + 616 = 2296$ m^2 (1).	1+1+1
31	$P(\text{prime}) = \frac{52}{100} = \frac{13}{25}$; $P(>4) = \frac{31}{100} = \frac{31}{100}$ (1); Either $= 52 + 31 - 14 = 69$ (1); $P = \frac{69}{100} = \frac{69}{100}$ (1).	1+1+1
	OR: Terms: $12, 17, 22, \ldots, 197$ (1); Number of terms $= 38$ (1); Sum $= \frac{n}{2}(a + l) = 3971$ (1).	1+1+1
32	$x = 12$ is a multiple of 3; $y = 60 = 12 \times 5$ (1); General: $x = ap$, $y = xq = a(pq)$ (1); A multiple of the given number (1).	1+1+1
	OR: Weighted sum $= 80 \times 4 + 78 \times 3 + 54 \times 8 + 85 \times 2 = 1156$ (1); Sum of weights $= 17$ (1); Weighted average $=$ 68 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) The perpendicular from the centre bisects the chord, so the half-length is 12 m (1). (ii) $\sqrt{13^2 - 12^2} = 5$ m (1). (iii) Half-chord $= \sqrt{13^2 - 12^2} = 5$, so the chord is 10 m (1). (iv) Distances from the centre are 5 m and 12 m on opposite sides, so the distance between them is $5 + 12 = 17$ m (2).	1+1+1+2
34	(i) $2\pi r = 2 \times \frac{22}{7} \times 14 = 88$ m (1). (ii) $\pi r^2 = \frac{22}{7} \times 196 = 616$ m^2 (1). (iii) Outer radius $= 21$ m, area $= \frac{22}{7} \times 441 = 1386$ m^2 (1). (iv) Path area $= 1386 - 616 = 770$ m^2 (1); cost $= 770 \times 40 =$ Rs 30800 (1).	1+1+1+2
35	Let x be removed: $\frac{12-x}{27-x} = \frac{1}{4}$ (1); $4(12 - x) = 1(27 - x)$ (1); $48 - 4x = 27 - x$ (1); $x = 7$ (2).	1+1+1+2
36	Ratio $\frac{2}{3}$ is constant, so a GP (1); 1st rebound $= 54$ m (1); 3rd rebound $= h \times (2/3)^3$ (1); $= 24$ m (2).	1+1+1+2
	OR: $9^2 = 81 = 10 \times 8 + 1$ (1); Let the odd integer be $2m + 1$ (1); Square $= 4m^2 + 4m + 1 = 4m(m + 1) + 1$ (1); $m(m + 1)$ is even, so $4m(m + 1)$ is a multiple of 8 (1); Hence $8k + 1$ (1).	1+1+1+1+1
37	January food $= 1360$ (1); February food $= 1292$ (1); January is higher by 68 (1); The larger share outweighs the smaller total in January (2).	1+1+1+2
	OR: gcd(882, 490): $(882, 490) \to (490, 392) \to (392, 98) \to (294, 98) \to (196, 98) \to (98, 98) \to (98, 0)$ giving 98 (1); gcd(98, 630): $(630, 98) \to (532, 98) \to (434, 98) \to (336, 98) \to (238, 98) \to (140, 98) \to (98, 42) \to (56, 42) \to (42, 14) \to (28, 14) \to (14, 14) \to (14, 0)$ (2); gcd of the three numbers $= 14$ (2).	1+2+2
38	Diagonals of a parallelogram bisect each other, so $OA = OC$ and $OB = OD$ (1); $OA^2 + OB^2 = 3600 + 3969 = 7569 = AB^2$ (1); By the converse of the Baudhāyana–Pythagoras theorem, $\angle AOB = 90°$, so the diagonals are perpendicular (1); A parallelogram with perpendicular diagonals is a rhombus, so all sides equal (1); Perimeter $= 4 \times 87 = 348$ cm (1).	1+1+1+1+1
	OR: Rate $= \dfrac{476 - 176}{14 - 2} = 25$ per km (1); Fixed charge $= 176 - 25 \times 2 = 126$ (1); Fare for 27 km $= 25 \times 27 + 126$ (1); $=$ ₹801 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.